

Science collaboration is space compass, not conflict

Original Editorial

LAST WEEK, NASA announced an **ambitious** roadmap for **sustained** operations on the Moon over the next decade. The agency reportedly **envisions** creating a base on the lunar South Pole in three parts: Getting to the moon's surface and beginning experiments from 2026 to 2029; "initial operations", including potential nuclear power installations in the next three years; and **commencing** work on a "semi-permanent" human station after 2032. The choice of location is significant. Unlike many regions of the Moon, where periods of sunlight and darkness alternate about every two weeks, parts of the South Pole experience shorter periods of shadow. They can thus provide solar power to drive the base's energy operations. At the same time, scientists believe that ice deposits in the region's permanently shadowed parts contain records of how water and other materials moved through the solar system. Studying these samples could help understand processes that created conditions for life.

The lunar outpost is also expected to be a staging ground for **expeditions** to Mars. According to NASA, it could provide the first "long-term opportunity to study human physiological adaptation in a reduced-gravity field before a deep space voyage". This **endeavour** is closely tied to geopolitics. Over the past decade, China has built a **robust** space programme that has landed rovers on the Moon and brought lunar samples to its laboratories. Beijing and Moscow have also announced plans to build a nuclear-powered international space station within 100 km of the lunar South Pole. In other words, unlike the largely **symbolic** space competition of the last century between America and the Soviet Union, the race to the Moon and beyond is about gaining early access to strategically significant regions outside the Earth.

NASA's plans assign a **prominent** role to Amazon founder Jeff Bezos's aerospace company, Blue Origin, which has been tasked to develop heavy-lift rockets, lunar landers and early infrastructure for the project. Although Elon Musk's SpaceX doesn't seem to have found a place in the programme announced last week, its partnership with NASA

remains central to America's space ambitions. The **convergence** of great-power **rivalry** and commercial interests could accelerate scientific progress. At the same time, there are concerns that the Cold War-era legal framework is **ill-equipped** to govern the changing dimensions of space exploration. There is, of course, little prospect of an **imminent confrontation** between the US-led Artemis programme and the Sino-Russian International Lunar Research Station. However, the emergence of competing legal regimes risks **fragmenting** scientific cooperation at a time when challenges such as climate change demand collaboration between nations. Mechanisms must be found to ensure that scientific advancement, rather than geopolitical competition, remains the principal driver of space exploration.

Article Summary

The editorial analyzes NASA's recently announced roadmap for sustained lunar operations over the next decade, which focuses on establishing a base at the Moon's South Pole. This strategically chosen location offers continuous solar power and essential ice deposits that could reveal insights into the solar system's origins. Beyond scientific discovery, this lunar base will act as a staging ground for future Mars missions. However, the passage highlights a shift from the symbolic space race of the Cold War to a modern geopolitical contest involving the US, commercial entities like Blue Origin, and the Sino-Russian alliance. The author warns that outdated space laws and increasing rivalries could hinder global scientific cooperation. The editorial concludes by emphasizing that scientific progress, not geopolitical dominance, must remain the true compass of space exploration.

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Editorial Anatomy Round-Up

1. The Vision: NASA's Lunar South Pole Roadmap



2. The Motivation: Strategic Access & Mars Prep



3. The Complication: Sino-Russian Rivalry & Outdated Laws



4. The Conclusion: Science Must Lead Over Politics

Vocabulary Builder: 15 Difficult & Important Words

Word (PoS)	Meaning (Hindi / English)	Synonyms (5)	Antonyms (5)
ambitious (Adjective)	महत्वाकांक्षी Having or showing a strong desire and determination to succeed.	aspiring, determined, driven, forceful, enterprising	lazy, unmotivated, easygoing, lethargic, unenthusiastic
sustained (Adjective)	निरंतर / लगातार Continuing for an extended period or without interruption.	continuous, prolonged, steady, unending, perpetual	intermittent, broken, fleeting, temporary, brief

Word (PoS)	Meaning (Hindi / English)	Synonyms (5)	Antonyms (5)
envisions (Verb)	कल्पना करना Imagine as a future possibility; visualize.	imagines, pictures, visualizes, anticipates, foresees	ignores, overlooks, neglects, disregards, blinds
commencing (Verb)	शुरू करना Begin; start.	beginning, starting, initiating, launching, opening	ending, concluding, finishing, terminating, halting
expeditions (Noun)	अभियान Journeys or voyages undertaken by a group of people with a particular purpose.	journeys, voyages, tours, excursions, missions	halts, stays, delays, blockages, stoppages
endeavour (Noun)	प्रयास / उद्यम An attempt to achieve a goal.	attempt, effort, venture, undertaking, trial	inactivity, passivity, idleness, laziness, abandonment
robust (Adjective)	मज़बूत / ठोस Strong and healthy; vigorous.	strong, vigorous, sturdy, tough, powerful	weak, fragile, feeble, frail, delicate
symbolic (Adjective)	प्रतीकात्मक Serving as a symbol.	representative, figurative, emblematic, token, indicative	literal, factual, actual, real, concrete
prominent (Adjective)	प्रमुख / प्रसिद्ध Important; famous.	important, well-known, outstanding, eminent, notable	obscure, unknown, hidden, invisible, insignificant
convergence (Noun)	मिलन / अभिसरण The process or state of converging (coming together).	merging, meeting, junction, intersection, confluence	divergence, separation, division, scattering, dispersion
rivalry (Noun)	प्रतिद्वंद्विता Competition for the same objective or for superiority.	competition, contention, conflict, opposition, clash	harmony, cooperation, alliance, partnership, friendship

Word (PoS)	Meaning (Hindi / English)	Synonyms (5)	Antonyms (5)
ill-equipped (Adjective)	अक्षम / अपर्याप्त रूप से सुसज्जित Not having the necessary resources or qualities for a particular role or task.	unprepared, unqualified, inadequate, unready, unfit	prepared, equipped, qualified, ready, capable
imminent (Adjective)	आसन्न / निकट About to happen.	impending, approaching, upcoming, forthcoming, looming	distant, remote, delayed, far-off, avoidable
confrontation (Noun)	टकराव / सामना A hostile or argumentative meeting or situation.	conflict, clash, fight, battle, encounter	agreement, peace, harmony, accord, compromise
fragmenting (Verb)	टुकड़ों में बटना Breaking or causing to break into fragments.	breaking, splintering, shattering, dividing, splitting	uniting, joining, combining, merging, assembling

Additional Important Words & Parts of Speech

- **Reportedly** — Adverb (According to what some say or what is reported).
- **Permanently** — Adverb (In a way that lasts or remains unchanged indefinitely).
- **Physiological** — Adjective (Relating to the branch of biology that deals with the normal functions of living organisms).
- **Strategic** — Adjective (Relating to the identification of long-term or overall aims and interests).
- **Mechanisms** — Noun (A system of parts working together in a machine; a process or system).

Idioms and Phrases

1. Staging ground

Meaning: A place where troops or people assemble before a major action or expedition.

Example: The military base served as a crucial staging ground for the upcoming overseas mission.

2. In other words

Meaning: Used to introduce an explanation of something just said, usually in a simpler or more explicit way.

Example: The project has been completely defunded; in other words, we have lost our jobs.

Practice Questions: Fill in the Blanks

1. The government announced an exceptionally _____ plan to overhaul the nation's infrastructure by the year 2030.

- A) modest
- B) apathetic
- C) grand
- D) indifferent
- E) unpretentious

2. Despite the recent economic downturn, the technology sector demonstrated a surprisingly _____ performance this quarter.

- A) frail
- B) delicate
- C) fragile
- D) sturdy
- E) brittle

3. The dark clouds and sudden drop in atmospheric pressure indicated that a severe thunderstorm was _____.

- A) distant
- B) receding
- C) delayed
- D) impending
- E) avoidable

4. The university administration is officially _____ its brand-new research program in renewable energy next semester.

- A) concluding
- B) initiating
- C) terminating
- D) pausing
- E) halting

5. The rapid growth of the modern tech industry is largely a result of the perfect _____ of technological innovation and market demand.

- A) divergence
- B) separation
- C) intersection
- D) isolation
- E) division

Practice Questions: Error Detection

6. (A) NASA announced an / (B) ambitious roadmap for / (C) sustain operations on the Moon / (D) over the next decade. / (E) No Error.

7. (A) The agency reportedly envisions / (B) creating a base on / (C) the lunar South Pole / (D) into three parts. / (E) No Error.

8. (A) Unlike many regions of the Moon, / (B) where periods of sunlight and darkness / (C) alternates about every two weeks, / (D) parts of the South Pole experience shorter periods of shadow. / (E) No Error.

9. (A) The convergence of great-power rivalry / (B) and commercial interests / (C) could accelerate / (D) scientific progress. / (E) No Error.

10. (A) Mechanisms must be found / (B) to ensure that scientific advancement, / (C) rather than geopolitical competition, / (D) remains the principal driver of space exploration. / (E) No Error.

Practice Questions: Reading Comprehension

Paragraph 1: NASA announced an ambitious roadmap for sustained operations on the Moon over the next decade. The agency envisions creating a base on the lunar South Pole in three parts: Getting to the moon's surface and beginning experiments from 2026 to 2029; "initial operations", including potential nuclear power installations; and commencing work on a "semi-permanent" human station after 2032. The choice of location is significant because parts of the South Pole experience shorter periods of shadow, allowing for solar power generation. Furthermore, scientists believe ice deposits in permanently shadowed parts contain records of how water moved through the solar system, providing clues to processes that created conditions for life. This lunar outpost is also expected to be a staging ground for expeditions to Mars.

Paragraph 2: This endeavour is closely tied to geopolitics. China has built a robust space programme and, alongside Moscow, announced plans to build an international space station near the lunar South Pole. Thus, the race is about gaining early access to strategically significant regions. The convergence of great-power rivalry and commercial interests (like Blue Origin) could accelerate scientific progress. However, there are concerns that the Cold War-era legal framework is ill-equipped to govern changing dimensions. The emergence of competing legal regimes risks fragmenting scientific cooperation when challenges like climate change demand collaboration. Mechanisms must ensure that scientific advancement, not geopolitical competition, remains the principal driver.

11. What is the primary reason for choosing the lunar South Pole for establishing a base?

- A) It contains a vast and unprecedented amount of gold deposits.
- B) It experiences shorter periods of shadow, enabling solar power, and contains ice deposits.
- C) It is the absolute closest point on the Moon to Earth for seamless communication.
- D) It is already populated by ancient alien infrastructure from a previous era.
- E) It allows for an easy and highly efficient launch path directly back to Earth.

12. How is the current race to space described as different from the Cold War space competition?

- A) It is now solely funded and controlled by private commercial billionaires.
- B) It involves only robotic exploration without any plans for human intervention.
- C) It is heavily focused on gaining early access to strategically significant regions.
- D) It is completely devoid of any underlying political or geopolitical motivations.
- E) It focuses entirely on establishing human colonies directly on the planet Mars.

13. According to the passage, what role does Blue Origin play in NASA's plans?

- A) It is solely tasked with providing funding for international astronauts.
- B) It is developing heavy-lift rockets, lunar landers, and early infrastructure.
- C) It is responsible for creating and governing the legal framework of space.
- D) It handles the public relations and global marketing for the lunar missions.
- E) It is actively building the Sino-Russian International Lunar Research Station.

14. What primary concern does the author raise regarding the current legal framework for space exploration?

- A) It is overly strict and completely prevents any commercial activity in outer space.
- B) It is outdated (Cold War-era) and ill-equipped to govern changing dimensions.
- C) It heavily favors the Sino-Russian alliance over the United States' Artemis program.
- D) It mandates that all new scientific discoveries must be kept entirely secret.
- E) It forces nations to focus exclusively on climate change instead of space exploration.

15. What is the ultimate message the author wishes to convey regarding space exploration?

- A) Intense geopolitical competition should be the sole driver to ensure rapid progress.
- B) Scientific advancement through collaboration must remain the principal driver.
- C) Private commercial companies should be banned from participating in lunar missions.
- D) All space exploration must be halted indefinitely until climate change is resolved.
- E) The Moon's resources should be divided equally among all nations on Earth immediately.

Answers & Explanations

Fill in the Blanks Answers

1. Answer: C) grand

Explanation: The original word in the editorial is "ambitious". "Grand" is the most appropriate synonym among the choices that fits the context of a large-scale plan.

2. Answer: D) sturdy

Explanation: The original word is "robust". "Sturdy" is a synonym that captures the meaning of strong and resilient performance.

3. Answer: D) impending

Explanation: The original text uses "imminent". "Impending" correctly signifies something that is about to happen very soon.

4. Answer: B) initiating

Explanation: The editorial uses "commencing". "Initiating" means to begin or start, which aligns perfectly with the sentence.

5. Answer: C) intersection

Explanation: The original sentence uses "convergence", meaning a meeting or coming together. "Intersection" effectively conveys this shared crossing point.

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Error Detection Answers

6. Answer: (C)

Explanation: The error is in part (C). The correct form of the adjective required here is "sustained", not the base verb "sustain", to properly modify the noun "operations".

7. Answer: (D)

Explanation: The error lies in part (D). The correct preposition to use when dividing or creating something composed of segments is "in". It should be "in three parts", not "into three parts" in this specific phrasing.

8. Answer: (C)

Explanation: The error is in part (C). The subject "periods" is plural. Therefore, the verb must also be plural. It should be "alternate", not "alternates".

9. Answer: (E)

Explanation: There is no grammatical or structural error in this sentence. The sentence is absolutely correct.

10. Answer: (C)

Explanation: The error is in part (C). The comparative conjunction used to show preference or alternative is "than", not the time-related adverb "then". It should be "rather than".

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Reading Comprehension Answers

11. Answer: B) It experiences shorter periods of shadow, enabling solar power, and contains ice deposits.

Explanation: Paragraph 1 explicitly states that the South Pole experiences shorter periods of shadow allowing for solar power generation, and contains ice deposits that could explain the solar system's origins.

12. Answer: C) It is heavily focused on gaining early access to strategically significant regions.

Explanation: Paragraph 2 notes that unlike the symbolic space competition of the past century, the current race is about "gaining early access to strategically significant regions."

13. Answer: B) It is developing heavy-lift rockets, lunar landers, and early infrastructure.

Explanation: Paragraph 2 explicitly mentions that Blue Origin "has been tasked to develop heavy-lift rockets, lunar landers and early infrastructure for the project."

14. Answer: B) It is outdated (Cold War-era) and ill-equipped to govern changing dimensions.

Explanation: The passage states there are concerns that the "Cold War-era legal framework is ill-equipped to govern the changing dimensions of space exploration."

15. Answer: B) Scientific advancement through collaboration must remain the principal driver.

Explanation: The final sentence of the passage clearly concludes that mechanisms must ensure that "scientific advancement, rather than geopolitical competition, remains the principal driver of space exploration."

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"The universe is under no obligation to make sense to you. But we are under an obligation to keep learning." – Neil deGrasse Tyson

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